europass Curriculum vitae

PERSONAL INFORMATION

Gianluca Peri

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www.gianlucaperi.com

www.github.com/gianluca-peri

POSITION

PhD Student in Machine Learning and Complex Systems' Physics

EDUCATION AND TRAINING

2024 – present

PhD in Machine Learning and Complex Systems' Physics

University of Florence, Italy

- Supervisor: Prof. Duccio Fanelli
- Co-Supervisor: Prof. Francesco Piazza
- In collaboration with: Tecnolink S.r.l.
- Research Group: CSDC – Center for the Study of Complex Dynamics

2025 – 2026

Long Research Stay

University of Namur, Belgium

- Supervisor: Prof. Timoteo Carletti
- Project: Hypergraph Neural Networks & Random Walks on Loss Landscapes

2021 – 2024

Master's Degree in Theoretical Physics

University of Florence, Italy

- Field: Theoretical Physics of Complex Systems
- Supervisor: Prof. Duccio Fanelli
- Thesis: *Generalized Spectral Formalism for Neural Networks*
- Grade: 110/110

2017 – 2021

Bachelor's Degree in Physics and Astronomy

University of Florence, Italy

- Supervisor: Prof. Simone Landi
- Thesis: *Fluid Models of Supersonic Winds from Stellar Coronae*

2011 – 2016

Scientific High School Diploma

Liceo Scientifico A. M. Enriques Agnoletti, Sesto Fiorentino, Italy

WORK EXPERIENCE

2025 – 2026

Science Communicator

OpenLab – Center for Science Education and Outreach, Sesto Fiorentino, Italy

2025 – 2026

Academic Tutor

University of Florence, Italy

SUPERVISION EXPERIENCE

2026 – present

Bachelor's Degree Thesis – Physics and Astronomy

University of Florence, Italy

Student: Francesco Fagone – *Configuration Entropy as a Learning Measure*

2025 – present

Master's Degree Thesis – Physics of Complex Systems

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University of Florence, Italy
Student: Francesco D'Angelo – *Spectral LSTM for Financial Time Series*

PUBLICATIONS

2026 D. Febbe, D. Fanelli, G. Peri, T. Carletti, *Optimal Navigation on Simplicial Complexes*, Preprint. https://arxiv.org/abs/2607.29450

2026 G. Peri, D. Febbe, D. Fanelli, *Spectral Higher-Order Neural Networks Have Sharp Expressivity Bounds*, Preprint. https://arxiv.org/abs/2607.19042

2026 G. Peri, T. Carletti, D. Fanelli, D. Febbe, *Spectral Higher-Order Neural Networks*, Under review. https://arxiv.org/abs/2603.28420

2026 G. Peri, L. Buffoni, G. Chiti, D. Fanelli, R. Marino, A. Nocentini, P. Panti, *Smart Walkers in Discrete Space*, Published in Chaos, Solitons & Fractals. https://doi.org/10.1016/j.chaos.2026.118356

2025 G. Peri, L. Chicchi, D. Fanelli, L. Giambagli, *SPectral ARchiteCture Search for neural network models*, npj Artificial Intelligence. https://doi.org/10.1038/s44387-025-00039-1

CONFERENCE AND WORKSHOP CONTRIBUTIONS

09/2026 **Talk – SIF 2026, 112th National Congress of the Italian Physical Society, Florence**
Title: *Spectral Higher-Order Neural Networks on Time Series*

06/2026 **Talk – SIFS, Italian Society of Statistical Physics, Parma**
Title: *Spectral Higher-Order Neural Networks*

06/2026 **Poster – Statistical Physics of Deep Learning School, Como**
Title: *Spectral Higher-Order Neural Networks*

05/2026 **Poster – PhD Day 2026, University of Florence, Florence**
Title: *Spectral Higher-Order Neural Networks*
Award: Poster Prize

04/2026 **Talk – NaXys Research Day, Namur**
Title: *Spectral Higher-Order Neural Networks*

12/2025 **Poster – BeNet 2025, 11th Belgian Network Research Meeting, Ghent**
Title: *Spectral Architecture Search for Neural Networks*

07/2025 **Talk – Nordita, Fluctuations in Self-Interacting and Learning Processes, Stockholm**
Title: *Training Walkers with Reinforcement Learning*

07/2025 **Poster – StatPhys29, 29th International Conference on Statistical Physics, Florence**
Title: *Spectral Architecture Search for Neural Networks*

07/2024 **Talk – COMPENG 2024, IEEE Workshop on Complexity in Engineering, Florence**
Title: *Generalized Spectral Formalism for Neural Networks*

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PERSONAL SKILLS

Mother tongue: Italian

Other languages

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	C1	C1	C1	C1	C1
French	A2	A2	A2	A2	A2

Levels: A1 and A2: Basic user – B1 and B2: Independent user – C1 and C2: Proficient user
Common European Framework of Reference for Languages

Operating systems: Linux, Windows

Software and programming languages: Python, PyTorch, Matplotlib, Jupyter, Git, LaTeX, bash, Visual Studio Code, Vim, C (basic), C++ (basic)

Other skills: Supervised and unsupervised learning, reinforcement learning, data analysis, numerical simulations, data mining, scientific computing, high-performance computing, software development, version control, Google Colab, HPC Clusters

REFERENCES

Prof. Duccio Fanelli – Full Professor, University of Florence
Prof. Francesco Piazza – Associate Professor, University of Florence
Prof. Timoteo Carletti – Full Professor, University of Namur

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